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Urban forest management in Addis Ababa, Ethiopia. A policy policy Arrangement Approach

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ABSTRACT

Recently in Ethiopia, forest decline has been observed in every part of the country and identified as an important problem. However, studies on policy approaches are scarce, and thus policy change analysis is imperative for understanding urban forest management. This study aims at analyzing Ethiopian forest policies using a Policy Arrangement Approach (PAA) with four dimensions: discourse, actors, power and resources, and rules of the game. Policy changes during three regimes (Imperial, 1936–1974; Derg, 1974–1991; and the present government, 1991 to present) are analyzed. A qualitative research design was employed to obtain and analyze data. Our analysis has revealed that policy changes from government to governance with the government. This major turn reflects a fundamental shift in the dominant discourse about the management of forests. It is found that PAA is a useful analytical tool to understand and explain policy changes. Insights from this analysis can contribute to the design of an integrated urban forest policy and critically reflect the challenges and interventions needed to positively influence forest management.

1. Introduction

The urban forest is a network of all woodlands, groups of trees, and individual trees located in urban and peri-urban areas (Sabatier, 1998) that provide ecosystem services. Mincey et al. (2013) indicate the human-engineered nature of urban forest management is determined by the actors and policy-makers to influence urban Planning, land use, and development. From the perspective of the integrative approach, Konijnendijk et al. (2005, p.20) viewed urban forestry as a series of strategic, interdisciplinary, and participatory approaches aimed at optimizing the Planning and management of urban green structures to provide multiple benefits to urban societies. Sandberg et al. (2015) consider three natures of urban forests (1) integrative, (2) socially inclusive, and (3) strategic which are important for urban forestry to develop as a collaborative approach. In the context of Addis Ababa, (Addis Ababa City Plan Preparation Program Office (AACPPPO), 2019) describes forests as a network of multi-purpose green spaces that provide ecosystem functions. For the purpose of this research, urban forest refers to all woodlands situated in urban areas and is delineated as urban forests. Urban forestry is generally defined as the art, science, and technology of managing trees and

forest resources in and around urban community ecosystems for the physiological, sociological, economic aesthetic Marutisthetic benefits trees provide to society (Harris et al., 2004).

The urban forest provides ecological, environmental, economic, and sociocultural benefits to humans (Wu, 2008). The ecological benefits of the urban forest include the ecosystem function and services. It contributes toward landscape connectivity through corridors, networks, and patches which form a nested spatial hierarchy (Carreiro et al., 2007); to conserve biodiversity and climate change mitigation (Barona et al., 2020). The environmental benefits can be explained in terms of improving air quality by absorbing particulates and pollutants (Wu, 2008); reducing building energy use (Nowak and Dwyer, 2007); The economic benefits are related to the monetary value of amenity benefits in recreation areas (Tyrvainen, 2001). The sociocultural benefits also include improving neighborhood and city landscape esthetics (Nowak and Dwyer, 2007); environmental education, and enhancing psychological and human well-being (Nowak and Dwyer, 2007; Peckham et al., 2013).

In Ethiopia, the first organized forest management intervention was started during the Italian annexation in 1936–41 (Melaku, 2003; Ayana,

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2014). However, it was not implemented. In 1962, Emperor Minelik II developed the first forest regulations intended to strengthen the national economy (Selassie, 1998). In 1975, the socialist government gave high political attention to forestry (Ayana, 2014) which led to the allocation of large tracts of national forest reserves (Kidane, 2002). Despite these efforts, deforestation and forest degradation continued (Melaku, 2003). Since the 1990s, improvements in forest management include the federal government decentralizing the responsibilities to regional governments, the participation of non-state actors in forest management, and the development of the Participatory Forest Management (PFM) issued in the 2007 forest law (Ayana, 2014; Ministry of Environment, Forest and Climate Change (MEFCC), 2018).

Ethiopia is one of the most rapidly urbanizing countries in Sub-Saharan Africa (Larsen et al., 2019). In 2007, the urban population of Ethiopia was about 11.9 million, this figure will increase to 42.4 million by 2037. According to the Central Statistical Agency of Ethiopia (CSA, 2012), Addis Ababa's population is estimated to reach about 3,434,000 (Central Statistical Agency, 2017). The total population of the city is also projected to reach 8,939,000 by 2037 (UN, 2014). This rapid urbanization has led to the depletion of natural resources. While the increasing urban populations rely on urban forests, which provide ecosystem services such as climate change mitigation and adaptation, watershed management, food and energy, health and wellbeing, biodiversity conservation, environmental quality, aesthetic, social, and ecological values (Konijnendijk et al. 2005; Tzoulas et al. 2007; Carreiro et al. 2007; Salbitano et al. 2016). Urban forests in Addis Ababa has been declining.

FDRE (2016) noted that "...an estimated 70,000 ha/yr of forest will be lost between 2010 and 2030" and over 3 million hectares of degraded forest land are under area enclosure (Lemenih and Kassa, 2014). In Addis Ababa, Horst (2006) shows natural forests remain only in small pockets around churches and on embassy compounds, covering an estimated area of about 250 ha only. Abebe and Megento (2016) show the loss of forests with a net change of 24.4 % from 2000 to 2015; and 16.66 % from 1986 to 2019 (Deribew, 2020). Studies on green space, in general, are skewed towards developed countries (Mensah, 2014). However, similar studies are neglected in third-world countries (Cilliers, 2010).

In Addis Ababa, the capital city of Ethiopia, studies on the urban forests are limited. Horst (2006) studied the rehabilitation of urban forests; Fetene and Worku (2013) investigated the Planning for the conservation and sustainable use of urban forestry; Woldegerima (2016) explained the implications of the urban environment and ecosystem services; Deribew (2020) focused on the spatiotemporal analysis of urban growth on forest and agricultural land. While several factors have contributed to the depletion of the urban forests, the fundamental problems are broadly related to institutions and policy issues. However, only a few studies, (Fetene and Hailu, 2013; Horst, 2006) show the effect of the failure of a strict conservationist approach, lack of urban forest policy, and forest law on the management of forestry.

Apart from these studies, a study on urban forest policy based on policy arrangements is relatively scant. This research aims to understand and analyze change and stability within forest policy arrangements and the challenges related to the present forest management. The research questions were: Which institutions are involved in policymaking and implementation? In what ways actors cooperate at different levels? And what are the challenges of forest management? It, therefore, provides new perspectives on the understanding of recent changes in forest policy and also proves to help improve those policies. This research sheds light on knowledge gaps and contributes to the emerging literature on urban forest policy. The analysis can also contribute to improving policies and their implementation for the future development of the urban forest.

2. Theoretical framework: Policy Arrangement Approach

In this research, an institutional type of policy analysis is used. This excludes the use of models like policy networks, interpretative policy

analysis, normative theory, and rational choice approaches (Veenman et al., 2009). Two institutional approaches are relevant for this: The Advocacy Coalition Framework/ACF (Sabatier, 1998) and the Policy Arrangement Approach or PAA (Arts et al., 2006). ACF considers the 'belief systems' of policy coalitions as the ultimate key factors for explaining stability, change, cooperation, and conflict, whereas the PAA takes a broader departure and distinguishes four key variables (Veenman et al., 2009). Among the variables, the concept of 'discourse', instead of 'belief system', a less individualistic and more dynamic ontology of policy practices is taken as a starting point (Hajer, 1995). In light of this, a Policy Arrangement Approach (PAA) was applied to analyze urban forest policies for Addis Ababa.

This approach was selected as an analytical framework because a Policy Arrangement Approach is a specific method of policy analysis that examines policy practice changes and structural changes in contemporary society (Park, 2015); it has only recently been developed to study policy dynamics in the environmental field (de Boer, 2006). It has been applied to analyze stability and change in the environmental policy domain (Janssen et al., 2015); it helps to understand the relationship between policy processes in which actors interact with structural forces of social and political change (Van der Zouwen, 2006).

PAA also addresses agency, structure, interests, and ideas in a dynamic perspective (Arts, 2006; Arts and Buizer, 2009). Further, PAA is a suitable analytical tool in various policy fields, including environmental policy, rural development policy, natural resource, and forest policy (Arts and Buizer, 2009; Veenman et al., 2009). Moreover, PAA has been applied to analyze changes in forest policies such as in the Netherlands (Veenman et al., 2009), Belgium (Van Gossum et al., 2011), Korea (Park and Youn, 2013; Park, 2015), and these studies demonstrate, a PAA is an appropriate way to describe the dynamics of change of policies with four interwoven dimensions. Besides, the Policy Arrangement Approach is used to analyze multilevel governance practices (Van der Zouwen, 2006). Policy arrangement is defined as the temporary stabilization of the content and organization of a policy domain (Arts et al., 2006, p.97). It includes structures formed through processes, interactions among policy actors and formal and informal rules (Park, 2015). Concerning environmental governance, the structural properties of policy arrangements include the Resources, Rules, and Discourses (Leroy and Arts, 2006). While Actors (the fourth policy dimension) will try to advance their interest by mobilizing their resources, make use of rules, frame discourses, or form coalitions (Arts et al., 2006).

Dynamic change and stability in a policy arrangement are influenced not only by strategic actions and interactions of actors but also by structural processes of social and political change or political modernization (Arts et al., 2006). Political modernization refers to structural processes that affect a certain governance mode (Arnouts et al., 2012). Within the field of environmental policy-making, political modernization refers to the shifting relations between state, market, and civic society (Tatenhove and Leroy, 2000). The word 'modernization' indicates the emergence of governance related to the involvement of citizens and other stakeholders in decision-making (Sandberg et al., 2015).

Regarding the dimensions, Arts and Buizer (2009) emphasize the importance of four interrelated analytical dimensions (discourses, actors, power, and rules) to understand policy practices. These four dimensions of a policy arrangement are inextricably interwoven (Park, 2015) and this relationship is symbolized by the tetrahedron (Fig. 1) in which each of the corners represents one dimension (Arts et al., 2006). A change in one of the dimensions will affect the other dimensions and change the shape of the entire figure (Fig. 1).

3. Methodology

3.1. Study area

Addis Ababa, the capital of Ethiopia, is the economic, political, and administrative hub of the country and the headquarter of the African

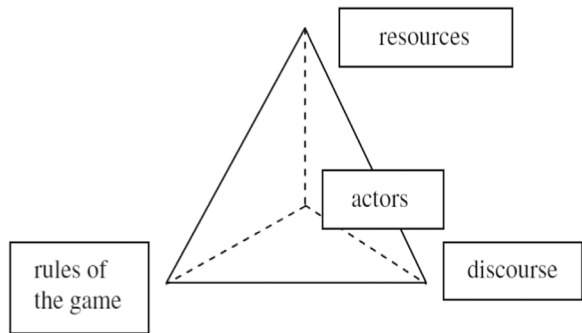


Fig. 1. The tetrahedron a symbol for the connections between the dimensions of an arrangement.
Source: Arts et al. (2006, p.99).

Union (AU), UN Economic Commission for Africa (UN-ECA), and a regional office for many international organizations (Ayele et al., 2021). The city is located at altitudes ranging from 2025 to 3028 m with a mean annual temperature range of 16–18 °C (Feyisa et al., 2014). The total population of Addis Ababa is estimated at 3,434,000 (Central Statistical Agency, 2007). The population is projected to increase by about 6 million in 2030 at an average annual growth rate of 4% (UNDESA, 2014). Its administrative hierarchy is structured into the two-tier arrangement, sub-cities, and *woredas* (small administrative unit) (Fig. 2).

3.2. Research methods

Primary and secondary data were employed in this research. Guided by four interrelated factors of a policy arrangement that consists of discourses, actors, power and resources, and rules. A comprehensive desk review of the literature related to legal, policy, and organizational issues related to forest planning and management was undertaken. This includes reports prepared by governments, organizations, and researchers on the area, and publications from various organizations’ websites and research articles. The main actor in forest management and Planning in the city involves government. Data collected from documents were categorized with the four dimensions of PAA and thematic content analysis was used to analyze the qualitative data. Criteria used to select documents include documents that convey information on forest management and Planning and documents relevant to green space and forest management efforts. The list (13) of reviewed documents can be found in (Table 1). Primary sources of data were obtained from 4 key

Table 1
Evolution of dominant discourses in Ethiopian forest policy, exemplified by main objectives of major policy documents in the field from 1994.

Year of issuing	Policy documents	Main objectives
1965	Proclamation No. 192/1965	Recognizes three forms of forest ownership: state, private, and protected forest
1980	Proclamation No. 192/1980	Recognizes state ownership of forests
1994	Forestry Proclamation No. 94/1994	Recognizes three forms of forest ownership: state, regional and private
1997	Forestry Proclamation No. 94/1997	Provides institutional, implementation, and legislative framework, responsibilities, and mandates for sustainable development of natural resources
1997	Environmental Policy of Ethiopia	Provides guidance and support in the conservation and sustainable utilization of Ethiopia’s environmental resources
2005	Urban Development Policy	Incorporate green spaces in national urban development spatial plans
2005	Ministry of Urban Development and Construction (MoUDC)	Developing green infrastructure standards and manuals for Ethiopian urban areas
2007	Forestry Proclamation No. 542/2007	To increase the forestry sector’s contribution to economic growth by protecting and developing the forest resource and fulfilling the rising demand for forestry products and services.
2009	CRGE (Climate Resilient Green Economy) strategy	To reduce greenhouse gas (GHG) emissions and increase climate resilience
	GTP (Growth and Transformation Plan) (2010/11–2014/15)	Focuses on Ethiopia’s green growth
2012	Guideline for Participatory Forest Management in Ethiopia	Facilitate the establishment and implementation of Participatory Forest Management (PFM)
2017	Addis Ababa City Planning Project Office (AACPO)	Developing the structure plan of Addis Ababa
2018	Proclamation No. 1065/2018	Establish private, community, association, and state forest ownership

informant interviews. The interviewees were selected based on their roles and experiences in the forest and related environmental governance issues in the city. The researcher interviewed key informants at the time of the interview in 2020. The interviewees included 2 from Environmental Protection and Green Development and 2 from Watersheds and Green Areas Development Agency (WGDA) offices. Each

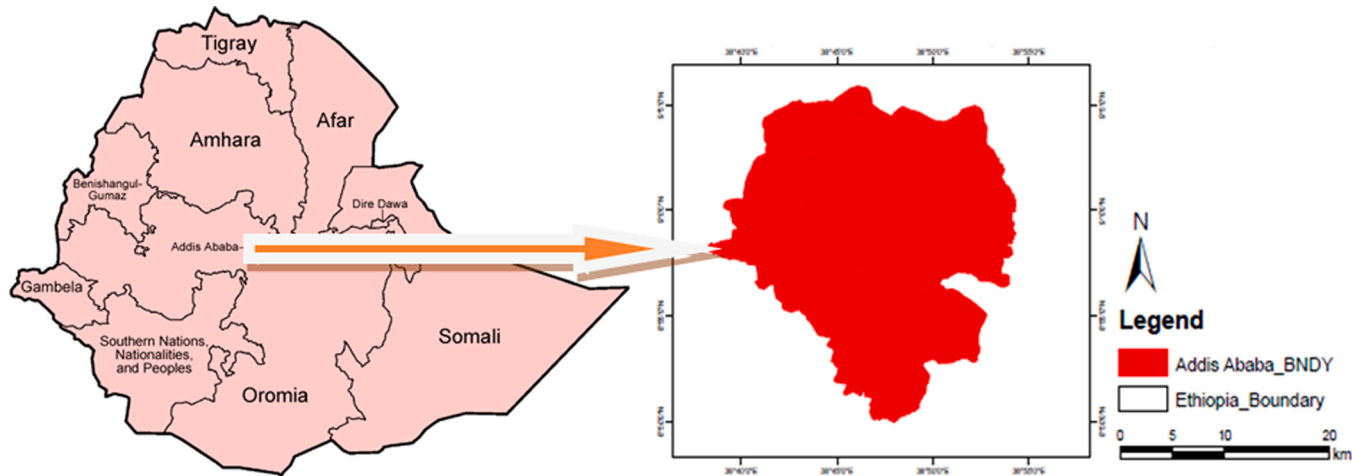


Fig. 2. Map of the study area.
Source: Ethio GIS (2021).

interview lasted between 60 and 90 min and was held in their offices. Data from interviews were transcribed, interpreted, and analyzed. We conducted an in-depth qualitative analysis of the historical process, the existing forest policy, and the challenges related to forest management.

4. Results

This research analyzes forest policies using a Policy Arrangement Approach (PAA) with four dimensions: discourse, actors, power and resources, and rules of the game. Policy changes by three regimes were analyzed; Imperial (1936–1974), Derg (1974–1991), and the present government (from 1991 to present). The analysis focuses on an examination of the interactions among the four dimensions of policy arrangement and the policies of the regimes.

4.1. Discourse

A discourse, the first dimension in the tetrahedron important to study the historical development of forest policy and institutions. Discourses in this context refer to the historical development of forest policies, legislation, and organizational setups of the forest sector in Ethiopia that have gone through many transformations. The first recorded forest management intervention began in 1936. The first modern legislation on forest resources was issued during the reign of Emperor Haile Selassie I (1930–1974) in 1965 (Nune, 2008).

In 1974, a military group known as Derg took power and declared socialism as its major philosophy guiding its political ideology and economic policies. It nationalized all rural and urban land and thereby eliminated the freehold land system. The Ethiopian Federal Democratic Republic Government (FDRE) overthrew the Derg regime in 1991 and introduced a free market economic system. Like the military regime, natural forests remained nationalized. The current government revised the 1975 proclamation in 1994 and 2007 with Forestry Proclamation No. 94/1994 and Forestry Proclamation No. 542/2007, respectively. Besides, the 1994 forest law recognized three forms of forest ownership: state, regional and private which led to the shift from a centralized to a multilevel arrangement system of forest governance (Negassa, 2014). In general, the focus of the three regimes on forest management shows that forest governance was embedded in the political economy of the country and dominantly elitist approach.

The current government adopted a new constitution in 1995 that heralded a decentralized federal polity and a democratic political process. The right to ownership of land and other natural resources, including forests, remained in the hands of the state and people (Article 40) and the government administers land on behalf of the people.

In Addis Ababa, the Eucalyptus species was introduced in 1895 (Horst, 2006). In the interest of safeguarding the supply of fuelwood, Menelik II introduced a range of eucalyptus forests in the vicinity of Addis Ababa and ordered the distribution of one hundred tree seedlings to each resident to plant, tend and utilize (Haile, 1991). Menelik II also ordered the residents to replant any areas that were so cleared (Ellis, 1988). From 1957 to 1964, the forest cover increased from 4000 to 10,400 ha. In addition, the state leased the land in and around Addis Ababa to the aristocrats and dignitaries (Haile, 1991). This, therefore, led to the proper management of forests before 1974. For example, Addis Ababa was referred to as 'Eucalyptopolis,' or the city of Eucalyptus trees, in 1925 (Horst, 2006).

This greenbelt was further expanded in the 1960s and 70s. The plantation area around Addis Ababa was about 13,500–15,000 ha in 1954 (FAO, 1985 in Bishaw, 2001; Von Breitenbach, 1962) and about 162,000 ha of plantation forest in 2001 (Bishaw, 2001). However, from 1974 onwards, the Eucalyptus plantations became subject to severe degradation (ULG Consultants Ltd., 1987) due to the absence of forest management; lack of precision in the delineation of local authority boundaries, and rising firewood demand from urban consumers. For example, within three years (1973–1976), plantations around Addis

Ababa decreased by about 33 % (Haile, 1991). While in 1980, proclamation No. 192/1980 allowed forest and wildlife conservation and development. Accordingly, to increase the supply of construction poles and fuelwood, plantation projects (for example, Addis-Bah Forestry Development Project) funded by African Development Bank (ADB), World Bank, and the government from the mid-1980s involved in planting 33,000 ha around Addis Ababa (Ethiopian Forestry Action Program (EFAP), 1994).

The forest policy issued in 2007 also introduced the concept of community participation with the intent to benefit the inclusion of communities residing inside and adjacent to a state forest. In 1980, the government developed forest and wildlife conservation and development Proclamation No. 192/1980 intended for the protection and management of forest and wildlife resources of the country. Further, the repealed forest Proclamation No.1065/2018 allowed state, private, community, and association forest ownership (provision of title deed), rights, and incentives. This indicates how the government perceived the benefits of forest resources for the country's development.

Several strategy and policy documents revealed government commitment toward environmental protection and a green economy in environmental and urban development policy. The Environmental Policy of Ethiopia (EPE) formulated in 1997, focused on sustainable natural resource development through the provision of a legislative framework, monitoring, and evaluation of environmental projects. This policy allowed the government, individuals, and associations to carry out forestry resource development and protection activities. Besides, incentives are considered to encourage investors and organizations engaged in forest resource development.

Likewise, the Climate-Resilient Green Economy Strategy (CRGE) of 2011 is the main policy driver for green growth in Ethiopia (Bass et al., 2013) aiming at the protection and reestablishing of forests for both economic and ecosystem benefits. CRGE Strategy is relevant for forest management in terms of green cities initiative for the creation of sustainable urban environments. In addition, Growth and Transformation Plan, GTP II (2015–2020) prescribes the importance of establishing and improving urban forest management in regional capital cities (Ministry of Environment, Forest and Climate Change (MEFCC), 2018) and increasing green area development and public recreation land utilization coverage to 30 % by 2020 (Ministry of Environment, Forest and Climate Change (MEFCC), 2018). The Environmental Protection Authority of Ethiopia (EPA, 2012) also focused on building a green economy.

With the influence of international discourse on urban forest policies, Ethiopia is a signatory of environmental conventions such as the Convention on Biological Diversity (CBD), the UN Convention to Combat Desertification (UNCCD), the UN Framework Convention on Climate Change (UNFCCC), and the Convention on International Trade in Endangered Species (CITES), and World Bank Forest Carbon Partnership Facility (FCPF) with REDD+ approach (Ministry of Environment, Forest and Climate Change (MEFCC), 2018). Among others, Agenda 21 is the one that recognizes the benefit of urban green resources for the urban poor by calling for the activation of "green works" programs (UN, 2003). For example, according to a key informant from WGDA, the Urban Productive Safety Net Project (UPSNP) supports the government in maintaining green spaces in the city. While the major influence of international discourse was on the rural forest.

Concerning planning, urban development policy (2005) focused on allocating green areas and mainstreaming environmental protection measures in all urban development activities. The city government stipulates the master plan of the city to incorporate green spaces. Among the ten master plans (1935–2017), the master plans developed between 1935 and 1941, and 1965 include a green belt and a green zone (Ayele et al., 2022a, 2022b). The 1986 master plan includes a city green system that consists of functional green areas reserved for recreational activities. The proportion of the 2002 master plan also constitutes 15 % of green structures (Ayele et al., 2022a, 2022b). However, the plans were

not implemented (ORAAMP, 2002; Mahiteme, 2007; Fetene and Worku, 2013; Woldegerima, 2016).

The urban forest in the city covers about 5000 ha of land categorized into dense, medium, and open forest strata (Woldegerima, 2016), and green space types in the city covers about 18,476 ha (Yeshitela, 2019). The present government has undertaken the Green Development Project, held by the Prime Minister intended to green part of the city's rivers and riversides to build some international recreational areas. Besides, 13 years ago the city government initiated the development of the Gullele Botanic Garden to conserve the flora of Ethiopia (Woldegerima, 2016). While studies identified the land use and land cover change of the urban forest. From 1957 to 1975 a net decline of 48.1% of forest land was identified due to the conversion of forest lands to agricultural lands (Deribew and Dalacho, 2019). About 1764 ha of forest was lost from 2003 to 2014 (Zewdie et al. 2018).

The changes in the land use of urban forests were shown a decrease from 12,168 ha (23.4%) to 10,301.5 ha (19.7%) which covers the city area in the year 2003–2016 (Azagew and Worku, 2020). In addition, the built-up areas within the forest area is increasing (Fetene and Worku, 2013; AACPO, 2017); the forest was cleared for new settlements and cropland (Deribew and Dalacho, 2019; Deribew, 2020); and the reserved land for the woodland forest in Akaki Kality and forest patches in Nifas-Silk sub-cities were converted into built-up areas (Azagew and Worku, 2020). Besides, the city's urban forest has significantly deteriorated due to the high rate of deforestation (Ministry of Environment, Forest and Climate Change (MEFCC), 2018).

The present, 2017 structure plan guides the city plan from 2013 to 2023. The plan also includes the conservation of forestry on sloppy areas and recreational park development on gentle slopes (AACPO, 2017). This plan identified access to functional recreational parks in the city as less than 1 m² per person. In this regard, the plan indicates the need to increase per capita green space by 9 m² per person in the future 25 years (AACPO, 2017) which is considered an ambitious plan (Yeshitela, 2019). This structure plan suggests reserving about 2730 ha of land that will be used for recreational parks. While, part of Adwa Park, a city park located near Bole airport partially parceled and redesigned for housing (Ayele et al., 2022a, 2022b) indicating gaps between the plan and its implementation.

Regulations No. 16/2004 allow the inclusion of environmental protection map that distinctly shows green places of parks, roads, and squares, forest places, green areas found on riversides, and agricultural places in the structural plan. In line with this, based on the experiences of international and national city's Urban Green Infrastructure (UGI) development and management, the Ministry of Urban Development and Construction (MoUDC) in collaboration with *Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)* has developed the Ethiopian national urban green infrastructure standard and green infrastructure management manuals considering the need to provide specific standards for location, capacity, and accessibility of urban green infrastructure in 2015.

Perhaps the stepwise change in the Planning is the inclusion of a multi-functional forest in the structure plan in 2017. From the above explanation, the evolution of forest policy over the past three regimes is characterized by a focus on different kinds of forest: from production forest (dominantly to supply of fuelwood in 1925, the 1960s and the 1970s, and plantation of trees from 1974 to 1991, to the inclusion of multi-functional forest in the plan and inclusion of non-state actors). This major turn reflects a fundamental shift in the dominant discourse about the management of forests.

4.2. Actors

The second dimension is actors. Analyzing a policy arrangement from an actor perspective requires the analysis of potential actors at different levels, as the forest is a multi-dimensional resource. The actors' roles were identified based on their position to cluster actors that fulfill

similar roles in the arrangement. Ethiopia is the Federal Democratic Republic with ten Regional States and two City administration councils. All regional states and city administrations have legislative, executive, and judiciary powers. In this regard, the FDRE Constitution recognizes the city of Addis Ababa as the capital city of the Federal State, and its residents shall have a full measure of self-government by being represented in the House of Peoples Representatives. Under the city government executive body is the city mayor, which constitutes the city manager, state functions, bureaus (cabinet), the mayor's office, and sub-city executive body. The city manager is in charge of municipal services. In this regard, currently, the Environmental Protection and Green Development Commission (EPGDC) is under the mayor's office. This organization has the power to monitor the Watersheds, Green Areas Development Agency (WGDA), and Gullele Botanical Garden Center. The forest management is one unit previously under (EPGDC) and shifted to (WGDA) Entoto and surrounding tourist destination unit.

At a federal/national level, the (MoUDC) and Ministry of Environment, Forest and Climate Change (MEFCC) now Environment, Forest and Climate Change Commission (EFCC) address the management of the forest. The former is responsible for developing urban planning legislation, policies, strategies, standards, and manuals. While the latter focuses on developing forest sector policies, strategies, proclamations, regulations, and guidelines. Accordingly, the government/state actors dominate the policy arrangement in which the forest resources policies are controlled by the government.

Perhaps significant changes in the government organization are the re-organization of executive organs. The Federal Environmental Protection Authority (EPA) was established in 1994 and re-established in 2002 as an independent institution by Proclamation no. 295/2002 (EPA, 2012), with a mandate to regulate and monitor all activities in the country that have an impact on the environment. In addition, the EPA was the leading governmental institution in developing and coordinating body for the Climate Resilient Green Economy (CRGE) strategy.

The Addis Ababa Environmental Protection Authority (AAEPA) was established at the city level in 1994. The Proclamation states the roles and responsibilities of the AAEPA. Concerning forest management, AAEPA was responsible for undertaking forest development activities in accordance with the master plan of the city and ensuring reserved forest areas under the holding of the city government are properly preserved. Proclamation No. 64/2019 stipulates re-establishing the environmental protection and green development commission and WGDA. Besides, the Addis Ababa City Government Rivers, Riversides Development and Climate Change Adaptation Project Office (ARDCAP), established by Regulation No. 75/2015, was again re-established by WGDA. Proclamation No. 64/2019 also provides relevant legal instruments, the definition of powers, and the duties of the city executive organs. A clear indication of this was the subsequent re-organizations in terms of organizing specific departments to carry out particular activities, including the natural resource research unit within WGDA, meaning that specialized expertise is an important resource in almost any policy arrangement. Another green development project under the prime minister's office that received political significance is called the *Sheger* project. This project focused on greening the city, particularly rivers and riverside development intended to build an international recreational standard riverbank including riverside forests.

At a city level, the Addis Ababa Environmental Protection Authority (AAEPA) was established in 1994. The Proclamation states the roles and responsibilities of the AAEPA. Concerning forest management, AAEPA was responsible to undertake forest development activities in accordance with the master plan of the city and ensuring reserved forest areas under the holding of the city government and are properly preserved. Proclamation No. 64/2019 stipulates the re-establishment of the environmental protection and green development commission and WGDA. Besides, the Addis Ababa City Government Rivers, Riversides Development and Climate Change Adaptation Project Office (ARDCAP), established by Regulation No. 75/2015, was again re-established to WGDA.

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In addition, the current forest management involves non-state actors and such actors are from the government (Addis Ababa university Space Science and *Sheger* greening); NGO (Resource initiative institute); and Orthodox Church. Among actors, the Orthodox Church has a basic respect for nature and emphasized the conservation of indigenous forests for ritual purposes, which in turn contributes to forest conservation ([Wondimu, 2007](#)). This shows the move from centralized (government) to governance by government. In line with this, the forest policy actors can be divided into (1) the government actors who interface with the resource because of legislation and management (MoUDC, MEFCC, EPGDC, WGDA); (2) research and academic institutions (Addis Ababa University); (3) non-governmental organizations (Resource initiative institute and urban productive safety net project); (4) religious organizations (Orthodox Church); and (5) forest resource users.

4.3. Power and resources

The third dimension entered into a tetrahedron is power and resources. The analysis of resource dependencies and power relations is based on traditional (neo-) Marxism or more recent approaches (multi-level governance) theoretical approaches ([Lieberink, 2006](#)). In this regard, actors around a given policy issue are to different degrees dependent upon each other for resources, e.g., money, information, or political legitimacy. In this case, the city government allocates the resources (budget) to different sectors, and the environment sector is the one. The resources and power relationship flows from federal (national) and regional (city) to the commission, WGDA, to the local level. Hence, the government holds the resources and is distributed them from the city to the local level through a top-down approach. Funds from NGOs are allocated for the implementation of a specific project. For example, the governmental structure includes a hierarchy of organs at the city, sub-city, and *woreda* (local) levels.

At a city level, the city government comprises the City Council, the Mayor, a City Cabinet, a Judicial Body, and the Chief Auditor. At the sub-city and *woreda* levels, the organs of government include the Council, Chief Executive, and a Standing Committee. As a chartered city and regional status, the government of Addis Ababa has the power in the management, planning, and administration of forest resources.

Accordingly, a range of actors often holds power; this division of power poses challenges as responsibilities can sometimes be unclear and decision-making can be slow. In the case of controlling informal settlements expansion, three tiers of government, the environmental protection commission, WGDA sub-cities, and *woredas* peace and security unit are responsible. While lack of cooperation was observed between the environmental protection commission and WGDA to assign forest guards for specific days that may create the possibilities for informal settlement expansion. For example, key informants from EPGDC had this to say: *a lack of coordination between the environmental protection commission and WGDA in assigning forest guards creates gaps and allows the invasion of the forest areas*. In addition, interviewed experts also mentioned that forest guards are not liable and they may engage in bribery that allows the informal settlement expansion. Besides, local-level positions are occupied by the politicians mainly responsible to maintain political stability. This may lead sometimes to negligence (intentionally or not) to take action. Particularly, *woreda* peace and security officials are responsible

to patrol and demolishing the informal settlements. According to interviewed sub-cities experts, the situation may create bribery. This in general is related to weak inventory, monitoring and evaluation mechanisms, limited cooperation between sectors, blurred power relationships among the commission, WGDA, and local levels, and lack of political commitment.

4.4. Rules of the game

The fourth dimension of a policy arrangement is the rules of the game. Rules are the mutually agreed formal procedures and informal routines of interaction within institutions ([Lieberink, 2006](#)). These rules guide policy formulation, decision-making, and the implementation of policies ([Arts and Buizer, 2009](#)). Rules of the game also refer to regulations, legislation, and procedures relevant to a certain policy domain ([Sandberg et al., 2014](#)) and rules of the game are institutions ([Van Bueren, 2009](#)). Further, [Lieberink \(2006\)](#) describes rules as a suitable strategy for studying the influence of institutional change on particular policy areas. In this regard, in Ethiopia, the federal government develops legal rules (environmental protection, urban Planning, management legislations, and proclamations) while the management of forests mainly operates at the city and local levels. The legal rules or policies provide a positive outlook but these are often not accompanied by sufficient implementation to realize legal or policy objectives.

The function of the city government includes generating revenues from land leases and as custodian of municipal services indicated in the structure plan. [Proclamation No. 574/2008](#) elucidates the protection of the natural environment and ensuring sustainable development. The Addis Ababa City Planning Project [AACPO \(2017\)](#) incorporates the multifunctional green space intended for the conservation and protection of forests and prohibits the construction of structures above ground. Forest ownership is vested in the government and the government institutions are responsible for the management and Planning. Further, it is stated that the city government shall take forest areas under private holdings. On contrary, the city government allows the construction of real estate called *Birhan-Gohe* real estate (key informant from environmental protection commission and WGDA) in the Yeka mountain forest that the city government reasserts its power and emphasizes the economic benefits of the land lease. Besides, the dual functions, to provide land on leasehold sales and approving the structure plan of the city may lead to dwindling or the conversion of forest areas. For example, *Jemmo* Mountain planned for a multi-functional forest but converted it to condominium housing; and an informal settlement expansion in forest areas. Besides, the recent reports ([Addis Ababa City Plan Preparation Program Office \(AACPO\), 2019](#)) indicate the expansion in the Yeka forest continues, by which all contributed to deforestation. The influence of rules and institutions on forest management shows an unstable pattern due to the gaps between policy arrangement and its implementation which negatively affect forest management in the city.

The evolution of forest policy over the past three regimes is characterized by a focus on different kinds of the forest: from production forest, plantation of trees, and to the inclusion of multi-functional forests in the plan. The changes in the government organizations were attributed to the re-organization of executive organs which needs specialized experts and the shift from the government as the only actor to the involvement of non-state actors. While the unclear and sometimes blurred division of power contributed to deforestation. The shift in the country's forest policy was embedded in the political economy of the country and emphasized a dominantly elitist approach.

5. Discussion

5.1. Discourse

A discourse helps to study the historical development of forest policy and institutions. The first recorded forest management intervention

began during the Italian occupation, 1936–41 (Bekele, 2003), focusing on forest inventory and exploitation. The Italians established a forest administration in 1937 (Negassa, 2014). From 1941–1974 the predominant policy was modernization that followed Western industrialized countries (Zewde, 1991). For example, the 1965 legislation on forest resources recognized three forms of forests (namely state forest, private forest, and protected forest) (Nune, 2008) intended to enlarge the sources of state revenue (Rahmato, 2001). During (1974–1991), all forests and land were nationalized and governed by hierarchies of commands (Negassa, 2014). Due to the 1984/5 famine, the government designed collectivization, villagization, and resettlement programs which had devastating effects on the forest resources of the country (Rahmato, 2001). In turn, the government has undertaken the plantation of forests mainly for commercial timber, poles, fuelwood, and construction timber (Million, 2011). Further, a legal contract under the 1994 forest law gave forest owners considerable freedom in their management decisions (Negassa, 2014).

5.2. Actors

Analyzing a policy arrangement from an actor's perspective starts by identifying the relevant actors and their influence on the policy process (Lieberink, 2006) because actors take part in governing activities (Kooiman, 2003). Besides, Krott (2005), Maryudi (2016) indicate the need to analyze potential actors from local to international levels. Since the forest is a multi-dimensional resource, in this case, the roles of actors were identified based on their position to cluster actors that fulfill similar roles in the arrangement. Accordingly, the federal line ministries are responsible for translating broader policy targets and objectives into their respective sectoral action plans. This includes the translation of cross-sectoral proclamations into action plans and the coordination of the respective regional state institutions for the implementation of action plans through financial disbursement and technical backup (Ministry of Environment, Forest and Climate Change (MEFCC), 2018).

Among actors, establishing a church forest program would be relevant to the future urban forest management by introducing a specific legal framework in the management of urban forests nest on the premises of the Orthodox Church. Because, the Orthodox church has a long history of conservation of forest resources (Wassie and Teketay, 2006). Church forests in Ethiopia, are mainly preserved for spiritual values which contribute to the restoration of degraded landscapes through integration into regional landscape planning and management policies (Sahle et al., 2021, p.1).

Establishing a church forest program would be important to future urban forest management by introducing a specific legal framework for the management of urban forests on the premises of the Orthodox Church because the Orthodox church has a long history of conservation of forest resources (Wassie and Teketay, 2006). Despite the contribution to forest management in rural areas, in urban areas, the church contributes to the management and maintenance of green spaces. Its contribution as an urban forest is stated in terms of maintaining indigenous trees. For example, about 60 % of the indigenous plant species were recorded in Hawassa urban church forests (Feyisa et al., 2022, p.6). Cutting trees in the church is considered taboo and a sign of a heaven garden (Wassie and Teketay, 2006; Feyisa et al., 2022).

5.3. Power and resources

The management of forests is also the responsibility of three layers of government (national, regional/city, and local level). Indicating a range of actors often holds power. This division of power poses challenges because responsibilities can occasionally be unclear and decisions might be taken slowly. Three levels of government, the environmental protection commission, WGDA, sub-cities, and the *woredas* peace and security unit are in charge of limiting the growth of informal settlements. The lack of coordination between the Environmental Protection

Commission and WGDA on the deployment of forest guards, for instance, creates a gap and allows encroachment into forest areas. This is similar to Fetene and Hailu (2013), Horst (2006) noted that the squatters had invaded the forest areas and the need to rehabilitate the area. This is generally attributed to a lack of political commitment, a lack of a strong inventory, monitoring and evaluation systems, limited sectoral cooperation, and blurred power dynamics between the commission, WGDA, and local levels.

5.4. Rules of the game

Rules of the game refer to regulations, legislation, and procedures relevant to a certain policy domain (Sandberg et al., 2014). Gaps are observed in the legislation and management of the forest. It is stated that the city government shall take forest areas under private holdings. On the contrary, the city government allows the construction of real estate in the Yeka mountain forest. The influence of rules and institutions on forest management shows an unstable pattern due to the gaps between policy arrangement and implementation, which negatively affect forest management in the city.

The interconnections between the four policy arrangement aspects and the policies reveal how discourse, the role of actors, their power relations, resources, institutional reorganization, and rules and institutions have positive and negative implications for forest management and development. With four interconnected dimensions, PAA is described as a useful analytical tool to describe the dynamics of the policy change and continuity with four interwoven dimensions (Arts and Buizer, 2009; Ayana et al., 2013; Park, 2015).

6. Conclusion

In this research, we have used a policy arrangements approach to understand successive changes in Ethiopia's forest policy since 1936. The result shows that the historical development and the shift in the country's forest policy were embedded in the political economy of the country and emphasized a dominantly elitist approach. The forest policy is formulated at a national level with a focus on rural forests and there is no specific urban forest policy. The government actors dominate the policy arrangement. In Addis Ababa, forests were predominantly used for fuelwood and construction materials. Changes in forest management show the move from government to governance by the government, focus on the green economy, and the provision of forest ownership title deeds to state, private, community, and associations that contributes for forest management and development. However, in the study area, this change is not without challenges. It is difficult to mention the existence of proper management of forests in the city. Because of the present political situation, limited cooperation between organizations, unclear responsibilities, blurred power relationship, lack of check and balance, less political commitment, and gaps in policy arrangement and Planning and its implementation creates pressure on the management of forests. This in general requires an integrative approach; the participation of professionals, NGOs, religious organizations, civil society and community-based organizations, and the users with the government actors. It is suggested that introduced a specific legal framework for the management of urban forests and forests on the premises of the Orthodox Church; introducing an urban forest policy that focuses on the social bargaining process for the regulation of conflict of interest in the protection of forests, and establishing urban forest programs through long term development and management are necessary. This research revealed that PAA is relevant to understanding and analyzing policy changes in urban forest management.

CRedit authorship contribution statement

Contribution: all research processes (conceptualizing, literature search, figures, study design, data collection, data analysis, data

interpretation, writing etc.) **Kumelachew Yeshitela Habetemariam**: Main supervisor. Contribution: revising the manuscript critically for important intellectual content. **Tebarek Lika Megento**: second or co-supervisor, revising the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- AACPO, 2017. The Structure Plan of Addis Ababa. Addis Ababa City Planning Project Office.
- Abebe, M.T., Megento, T.L., 2016. The city of Addis Ababa from 'forest City' to 'urban Heat Island' assessment of urban green space dynamics. *J. Urban Environ. Eng.* 10 (2), 254–262.
- Addis Ababa City Plan Preparation Program Office (AACPPPO), 2019. A Report on Informal Expansion. Addis Ababa, Ethiopia.
- Arnouts, R., van der Zouwen, M., Arts, B., 2012. Analysing governance modes and shifts—governance arrangements in Dutch nature policy. *For. Policy Econ.* 16, 43–50.
- Arts, B., Buizer, M., 2009. Forests, discourses, institutions: a discursive-institutional analysis of global forest governance. *For. Policy Econ.* 11 (5–6), 340–347.
- Arts, B., Leroy, P., Van Tatenhove, J., 2006. Political modernization and policy arrangements: a framework for understanding environmental policy change. *Public Organ. Rev.* 6 (2), 93–106.
- Arts, B.J.M., 2006. Forest, institutions, discourses: a discursive-institutional analysis of global forest politics.
- Ayana, A.N., 2014. Forest Governance Dynamics in Ethiopia: Histories, Arrangements, and Practices. Wageningen University and Research.
- Ayana, A.N., Arts, B., Wiersum, K.F., 2013. Historical development of forest policy in Ethiopia: trends of institutionalization and deinstitutionalization. *Land Use Policy* 32, 186–196.
- Ayele, B.Y., Megento, T.L., Habetemariam, K.Y., 2021. Governance of green infrastructure planning in Addis Ababa, Ethiopia. *Land Use Policy* 111, 105777.
- Ayele, B.Y., Megento, T.L., Habetemariam, K.Y., 2022. Assessing green infrastructure spatial plans in Addis Ababa, Ethiopia. *Socio-Ecol. Pract. Res.* 1–17.
- Ayele, B.Y., Megento, T.L., Habetemariam, K.Y., 2022. The governance and management of green spaces in Addis Ababa, Ethiopia. *Heliyon* 8, 5.
- Azagew, S., Worku, H., 2020. Accessibility of urban green infrastructure in Addis-Ababa city, Ethiopia: current status and future challenge. *Environ. Syst. Res.* 9 (1), 1–20.
- Barona, C.O., Devisscher, T., Dobbs, C., Aguilar, L.O., Baptista, M.D., Navarro, N.M., da Silva Filho, D.F., Escobedo, F.J., 2020. Trends in urban forestry research in Latin America & the Caribbean: a systematic literature review and synthesis. *Urban For. Urban Green.* 47, 126544.
- Bass, S., Wang, S.S., Ferede, T., Fikreyesus, D., 2013. Making growth green and inclusive: the case of Ethiopia.
- Bekele, M., 2003. Forest property rights, the role of the state, and institutional exigency: the Ethiopian experience.
- Bishaw, B., 2001. Deforestation and land degradation in the Ethiopian highlands: a strategy for physical recovery. *Northeast Afr. Stud.* 7–25.
- de Boer, S., 2006. Diffusion or diversity in cultural heritage preservation? Comparing policy arrangements in Norway, Arizona and The Netherlands. In: *Institutional Dynamics in Environmental Governance*. Springer, Dordrecht, pp. 161–181.
- Carreiro, M.M., Song, Y.C., Wu, J., 2007. Ecology, Planning, and Management of Urban Forests: International Perspective. Springer Science & Business Media.
- Carreiro, M.M., Song, Y.C., Wu, J., 2007. Ecology, Planning, and Management of Urban Forests: International Perspective. Springer Science & Business Media.
- Central Statistical Agency, 2007. Population and Housing Census Report-Country. Addis Ababa, Ethiopia.
- Central Statistical Agency, 2017. Statistical Abstract. Addis Ababa, Ethiopia.
- Cilliers, E.J., 2010. Future directions in Urban Planning and Space Usage. *Interdisciplinary Themes Journal* 1 (1).
- Deribew, K.T., 2020. Spatiotemporal analysis of urban growth on forest and agricultural land using geospatial techniques and Shannon entropy method in the satellite town of Ethiopia, the western fringe of Addis Ababa city. *Ecol. Process.* 9 (1), 1–16.
- Deribew, K.T., Dalacho, D.W., 2019. Land use and forest cover dynamics in the North-eastern Addis Ababa, central highlands of Ethiopia. *Environ. Syst. Res.* 8 (1), 1–18.
- Ellis, G., 1988. In search of a development paradigm: two tales of a city. *J. Mod. Afr. Stud.* 26 (4), 677–683.
- EPA, 2012. National Report of Ethiopia, the United Nations Conference on Sustainable Development (Rio+20). Federal Democratic Republic of Ethiopia, Addis Ababa.
- Ethiopian Forestry Action Program (EFAP), 1994. The Challenge for Development.
- FAO, 1985. *Tree Growing by Rural People*. Forestry Paper 64. FAO, Rome.
- Federal Democratic Republic of Ethiopia (FDRE), 2016. Ethiopia's Forest Reference Level Submission to the UNFCCC.
- Fetene, A., Worku, H., 2013. Planning for the conservation and sustainable use of urban forestry in Addis Ababa, Ethiopia. *Urban For. Urban Green.* 12 (3), 367–379.
- Feyisa, A., Negash, M., Melka, Y., 2022. Urban green infrastructure affects woody plant diversity and carbon stock in Hawassa city in Ethiopia. *Arboric. J.* 1–15.
- Feyisa, G.L., Dons, K., Meilby, H., 2014. Efficiency of parks in mitigating urban heat island effect: an example from Addis Ababa. *Landsc. Urban Plan.* 123, 87–95.
- Haile, F., 1991. Women Fuel Wood Carriers in Addis Ababa and the Peri-Urban Forest: Report to International Development Research Centre (IDRC) and National Urban Planning Institute (NUPI). International Labour Office, Geneva, CH.
- Hajer, M.A., 1995. The politics of environmental discourse: ecological modernization and the policy process.
- Harris, R.W., Clark, J.R., Matheny, N.P., 2004. Arboriculture: integrated management of landscape trees. Shrubs Vines.
- Horst, A., 2006. Rehabilitation of urban forests in Addis Ababa. *J. Drylands* 1 (2), 108–117.
- Janssen, S.K., van Tatenhove, J.P., Otter, H.S., Mol, A.P., 2015. Greening flood protection—an interactive knowledge arrangement perspective. *J. Environ. Policy Plan.* 17 (3), 309–331.
- Kidane, M., 2002. Tropical secondary forest management in Africa: reality and perspectives. In: *Workshop on Tropical Secondary Forest Management in Africa*. Nairobi, Kenya, 2002.
- Konijnendijk, C.C., Nilsson, K., Randrup, T.B., Schipperijn, J., 2005. *Urban Forests and Trees: A Reference Book*. Springer Science & Business Media.
- Kooiman, J., 2003. Societal governance. In: *Demokratien in Europa*. VS Verlag für Sozialwissenschaften, Wiesbaden, pp. 229–250.
- Krott, M., 2005. *Forest Policy analysis*. Springer Science & Business Media.
- Larsen, L., Yeshitela, K., Mulatu, T., Seifu, S., Desta, H., 2019. The impact of rapid urbanization and public housing development on urban form and density in Addis Ababa, Ethiopia. *Land* 8 (4), 66.
- Lemenih, M., Kassa, H., 2014. Re-greening Ethiopia: history. challenges and lessons. *Forests* 5 (7), 1717–1730.
- Leroy, P., Arts, B., 2006. Institutional dynamics in environmental governance. In: *Institutional Dynamics in Environmental Governance*. Springer, Dordrecht, pp. 1–19.
- Liefferink, D., 2006. The dynamics of policy arrangements: turning round the tetrahedron. In: *Institutional Dynamics in Environmental Governance*. Springer, Dordrecht, pp. 45–68.
- Mahiteme, Y., 2007. Carrying the Burden of Long-term Ineffective Urban Planning. An Overview of Addis Ababa's Successive Master Plans and their Implications on the Growth of the City. Working papers on population and land-use change in central Ethiopia, vol. 7, pp. 1–34/.
- Maryudi, A., 2016. Choosing timber legality verification as a policy instrument to combat illegal logging in Indonesia. *For. Policy Econ.* 68, 99–104.
- Melaku, B., 2003. Forest Property Rights, the Role of the State, and Institutional Exigency: The Ethiopian Experience [PhD dissertation]. Swedish University of Agricultural Sciences, Uppsala, Sweden.
- Mensah, C.A., 2014. Urban green spaces in Africa: nature and challenges. *Int. J. Ecosyst* 4 (1), 1–11.
- Million, B., 2011. Forest plantations and woodlots in Ethiopia. *Afr. For. Forum* 1 (12), 11–15.
- Mincey, S.K., Hutten, M., Fischer, B.C., Evans, T.P., Stewart, S.I., Vogt, J.M., 2013. Structuring institutional analysis for urban ecosystems: a key to sustainable urban forest management. *Urban Ecosyst.* 16 (3), 553–571.
- Ministry of Environment, Forest and Climate Change (MEFCC), 2018. National Potential and Priority Maps for Tree-Based Landscape Restoration in Ethiopia (Version 0.0): Technical Report. Ministry of Environment, Forest and Climate Change, Addis Ababa.
- Negassa, A., 2014. Forest Governance Dynamics in Ethiopia Histories, Arrangements, and Practices (Doctoral dissertation). Wageningen University.
- Nowak, D.J., Dwyer, J.F., 2007. Understanding the benefits and costs of urban forest ecosystems. In: *Urban and Community Forestry in the Northeast*. Springer, Dordrecht, pp. 25–46.
- Nune, S., 2008. Ethiopian Government efforts to increase forest cover: a policy oriented discussion paper. In: *Policies to increase forest cover in Ethiopia*, Proceeding of Policy Workshop, Environment for Development—Environmental Economics Policy Forum for Ethiopia, Ethiopian Development Research Institute, Global Hotel, Addis Ababa.
- Office for the Revision of Addis Ababa Master Plan (ORAAMP), 2002.
- Park, M.S., 2015. Inter-Korean forest cooperation 1998–2012: a policy arrangement approach. *Sustainability* 7 (5), 5241–5259.
- Park, M.S., Youn, Y.C., 2013. Development of urban forest policy-making toward governance in the Republic of Korea. *Urban For. Urban Green.* 12 (3), 273–281.
- Peckham, S.C., Duinker, P.N., Ordóñez, C., 2013. Urban forest values in Canada: views of citizens in Calgary and Halifax. *Urban For. Urban Green.* 12 (2), 154–162.
- Proclamation No. 192/1980. A Proclamation to Provide for the Conservation and Development of the Forest and Wildlife Resources. Addis Ababa, Ethiopia.
- Proclamation no. 295/2002. A Proclamation for Environmental Protection Organs Establishment Proclamation. Addis Ababa, Ethiopia.
- Proclamation No. 542/200. A Proclamation to Provide for the Development, Conservation and Utilization of Forests. Addis Ababa, Ethiopia.
- Proclamation No. 574/2008. A Proclamation to provide Federal Urban Planning. Addis Ababa, Ethiopia.

- Proclamation No. 94/1994. A Proclamation for Forest and Wildlife Conservation and Development. Addis Ababa, Ethiopia.
- Proclamation No.1065/2018. A Proclamation for forest Development, Conservation, and Utilization. Addis Ababa, Ethiopia.
- Proclamation No.64/2019. A Proclamation for the re-establishment of environmental protection and green development commission. Addis Ababa, Ethiopia.
- Rahmato, D., 2001. Environmental change and state policy in Ethiopia: lessons from past experience. *Forum Soc. Stud.*
- Sabatier, P.A., 1998. The advocacy coalition framework: revisions and relevance for Europe. *J. Eur. Public Policy* 5 (1), 98–130.
- Sahle, M., Saito, O., Reynolds, T.W., 2021. Nature's contributions to people from church forests in a fragmented tropical landscape in southern Ethiopia. *Glob. Ecol. Conserv.* 28, e01671.
- Salbitano, F., Borelli, S., Conigliaro, M., Yujuan, C., 2016. Guidelines on Urban and Peri-urban Forestry. FAO.
- Sandberg, L.A., Bardekjian, A., Butt, S., 2014. Urban Forests, Trees, and Greenspace: A Political Ecology Perspective. Routledge.
- Selassie, E.M.W., 1998. The forest resources of Ethiopia past and present. *Walia* 1998 (19), 10–28.
- Tatenhove, J.V., Leroy, P., 2000. The institutionalization of environmental politics. In: *Political Modernization and the Environment*. Springer, Dordrecht, pp. 17–33.
- Tyrvaenen, L., 2001. Economic valuation of urban forest benefits in Finland. *J. Environ. Manag.* 62 (1), 75–92.
- Tzoulas, K., Korpela, K., Venn, S., Yli-Pelkonen, V., Kaźmierczak, A., Niemela, J., James, P., 2007. Promoting ecosystem and human health in urban areas using Green Infrastructure: a literature review. *Landsc. Urban Plan.* 81 (3), 167–178.
- ULG Consultants Ltd, 1987. Addis-Bah Forestry Development Project, Socio-economic Study. Final report, Vol. 1, summary.
- UN, 2014. World Urbanization Prospects: The 2014 Revision-Highlights. UN.
- United Nations Centre for Human Settlements (Habitat 2003). The Challenges of Slums.
- Van Bueren, E., 2009. Greening Governance: An Evolutionary Approach to Policy Making for A Sustainable Built Environment, 30. Ios Press.
- Van der Zouwen, M., 2006. Nature Policy between Trends and Traditions: Dynamics in Nature Policy Arrangements in the Yorkshire Dales, Doñana, and the Veluwe. Eburon Uitgeverij BV.
- Van Gossum, P., Arts, B., De Wulf, R., Verheyen, K., 2011. An institutional evaluation of sustainable forest management in Flanders. *Land Use Policy* 28 (1), 110–123.
- Veenman, S., Liefverink, D., Arts, B., 2009. A short history of Dutch forest policy: the 'de-institutionalisation' of a policy arrangement. *For. Policy Econ.* 11 (3), 202–208.
- Von Breitenbach, F., 1962. National forestry development planning: a feasibility and priority study on the example of Ethiopia. *Ethiop. For. Rev.* 3 (4), 41–68.
- Wassie, A., Teketay, D., 2006. Soil seed banks in church forests of northern Ethiopia: implications for the conservation of woody plants. *Flora-Morphol. Distrib. Funct. Ecol. Plants* 201 (1), 32–43.
- Woldegerima, T., 2016. Study of the Urban Environment and Ecosystem Services of Addis Ababa: Implications for Urban Greenspace Planning (Doctoral dissertation). Addis Ababa University.
- Wondimu, A., 2007. Which urban livelihood without adequate breathing space? A reflection on the green area of Addis Ababa. In: *First Green Forum Conference*, Addis Ababa, pp. 2–4.
- Wu, J., 2008. Toward a landscape ecology of cities: beyond buildings, trees, and urban forests. In: *Ecology, Planning, and Management of Urban forests*. Springer, New York, NY, pp. 10–28.
- Yeshitela, K., 2019. Urban green space development and management in a rapidly urbanizing sub-Saharan African city: the case of Addis Ababa, Ethiopia. *Ethiop. J. Environ. Stud. Manag.* 12, 1.
- Zewde, B., 1991. A History of Modern Ethiopia, 1855–1974. Ohio Univ Pr., pp. 1855–1974.
- Zewdie, M., Worku, H., Bantider, A., 2018. Temporal dynamics of the driving factors of urban landscape change of Addis Ababa during the past three decades. *Environ. Manag.* 61 (1), 132–146.